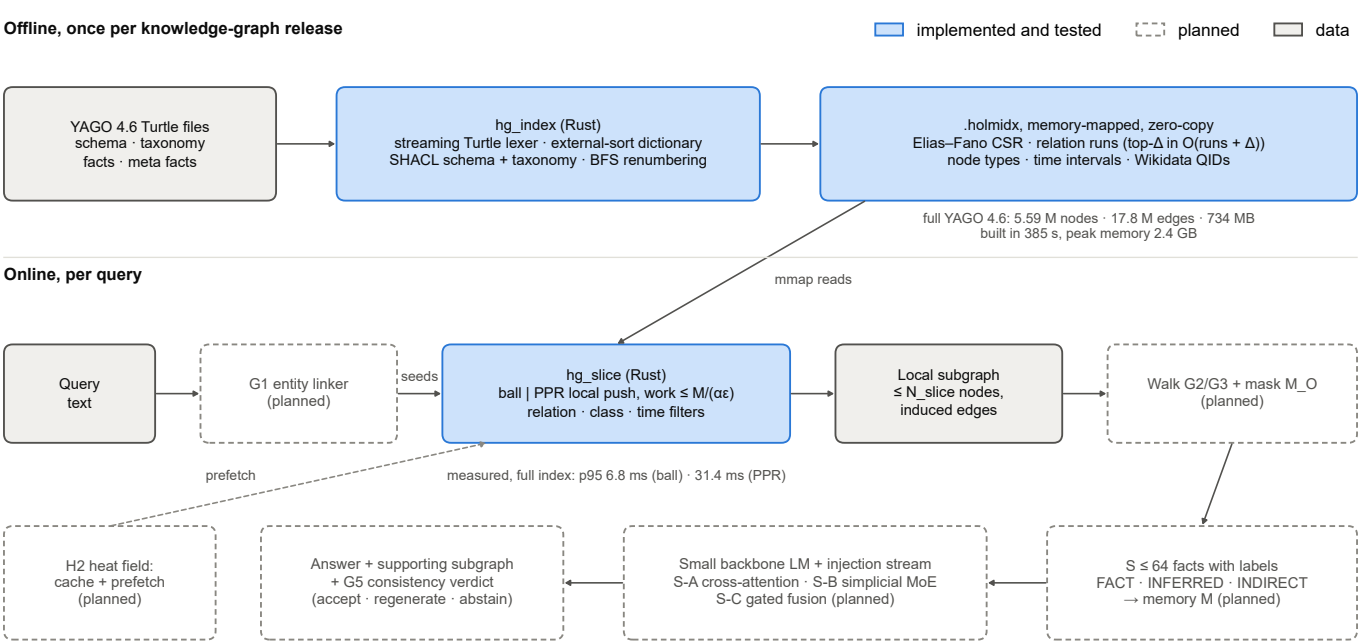
Offline, once per knowledge-graph release
implemented and tested
planned
data
YAGO 4.6 Turtle files
schema · taxonomy
facts · meta facts
hg_index (Rust)
streaming Turtle lexer · external-sort dictionary
SHACL schema + taxonomy · BFS renumbering
.holmidx, memory-mapped, zero-copy
Elias–Fano CSR · relation runs (top-Δ in O(runs + Δ))
node types · time intervals · Wikidata QIDs
full YAGO 4.6: 5.59 M nodes · 17.8 M edges · 734 MB
built in 385 s, peak memory 2.4 GB
Online, per query
mmap reads
Query
text
G1 entity linker
(planned)
seeds
hg_slice (Rust)
ball | PPR local push, work ≤ M/(αε)
relation · class · time filters
Local subgraph
≤ N_slice nodes,
induced edges
Walk G2/G3 + mask M_O
(planned)
prefetch
measured, full index: p95 6.8 ms (ball) · 31.4 ms (PPR)
H2 heat field:
cache + prefetch
(planned)
Answer + supporting subgraph
+ G5 consistency verdict
(accept · regenerate · abstain)
Small backbone LM + injection stream
S-A cross-attention · S-B simplicial MoE
S-C gated fusion (planned)
S ≤ 64 facts with labels
FACT · INFERRED · INDIRECT
→ memory M (planned)